# LT350-BELT SCALE

## Quick Reference Guide


REV 1.00

# INDEX

## LT350-BS BELT SCALE SYSTEM COMPONENTS

**LT350-BSI BELT SCALE INTEGRATOR**

**LT45-SS SPEED SENSOR WHEEL**

**LT45-BS WEIGH MODULE**

**WEIGH MODULE CARRIAGE**

| LT350-BSI BELT SCALE SPECIFICATION | |
|---|---|
| **LT350-BSI** | Belt scale weight integrator and belt speed input |
| | |
| **Belt Scale Functions** | Material Rate, Belt Speed (fixed or variable), Totalizer and statistical data logging plus a batch set point controller |
| **Accuracy** | ± 0.5 to 2% over 4 to 1 weigh idler span, e.g. 2 weigh idlers = 1.5% accuracy |
| **Belt Width** | Universal |
| **Belt Speed** | Dynamic speed up to 50 feet per second or static constant belt speed parameter |
| **Sampling Rate** | 1-500Hz, software selectable |
| **Span Stability** | 2.5ppm/ Celsius |
| **Zero Stability** | 6nV/Celsius |
| | |
| **Calibration Method** | Calibration through software menu system |
| | |
| **Filtering** | Programmable response filtering |
| **Firmware Upgrading** | In field Firmware upgrading using serial or Ethernet |
| **Unit Conversion** | Lb/kg/T/TN |
| **Display** | Full Graphics LCD display (White on Blue) |
| | |
| **Scale Input ranges** | 20mV range |
| **Load Cell Excitation** | 5VDC,8x350,16x700 ohm in total |
| | |
| **Serial Ports** | 1 full duplex RS232/RS485 |
| **Ethernet TCP/IP** | Ethernet 10/100 with PoE support – optional |
| **Wi-Fi TCP/IP** | 802.11b wireless support – optional |
| **USB** | Supports Virtual Serial Port interface - optional |
| | |
| **Power** | 12-48VDC 300mA nominal |
| **Temperature Range** | -10°C to +40°C (14°F to 104°F) |
| | |
| **LT350-BSI-xPSxx** | Stainless Steel Panel Mount |
| **LT350-BSI-xNDxx** | Aluminum NEMA4 wall mount (internal 110-240AC power supply included) |
| **Operating Class** | Class III/IIIL @10 000div |

## GENERAL CALIBTRATION SETUP REFERENCE

**DEFAULT FACTORY PASSWORD:**

There are close to a hundred parameters that can be set by either typing in the corresponding command number or simply by pressing the **[MENU]** key and navigating to the parameter of interest using the up/down arrow keys.

Protecting important settings from unauthorized tampering is a valuable feature and as such, most calibration and setup parameters are protected by a password. The password only needs to be entered once and only after the F1 key was pressed to save settings during the last session. Once the F1 key was pressed in calibration mode to save the new settings, calibration mode will be exited and to enter calibration mode again, the user needs to enter the password again.

The factory password is [1234]. It is recommended that the user change this password to a unique 4 number combination in order to protect the integrity of the system scale calibration and settings using **COMMAND[98]**.

For most commands the user only needs to enter calibration mode once by providing the password where required. 🔒

**LEGAL FOR TRADE APPLICATIONS:**

Some parameters are protected by a password – which is a government requirement for applications that will use the scale for trading to the public by weight. Such applications require an audit trail history of parameter changes and settings.

**NON LEGAL FOR TRADE APPLICATIONS:**

Many industrial applications are strictly used for process control such as batching and monitoring weight. These types of implementations do not require legal for trade certification or inspection by government agencies.

# IMPORTANT!!!

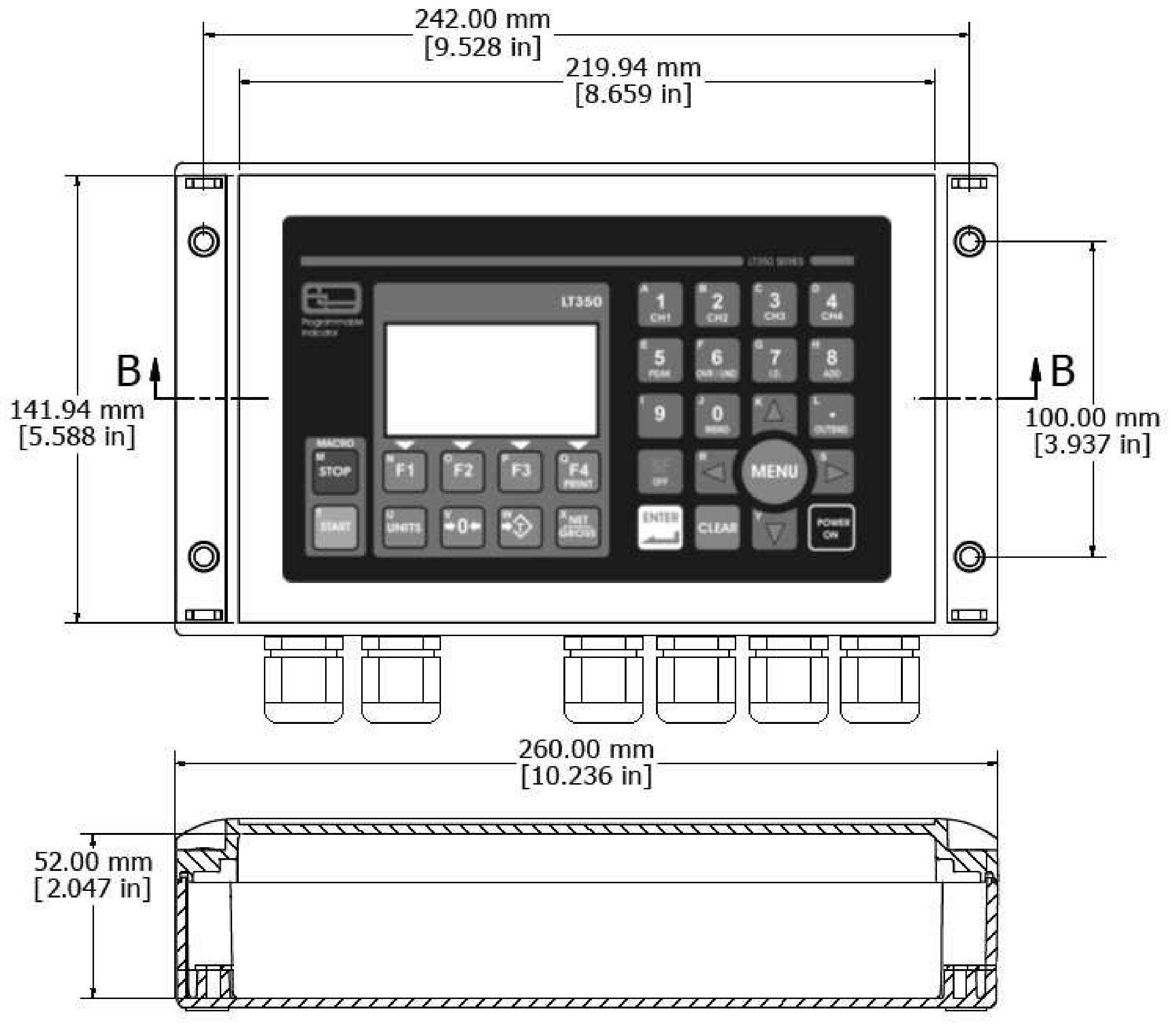
**It is important to note that no settings are saved in calibration mode until the user presses the F1 key**

## Scale Handling

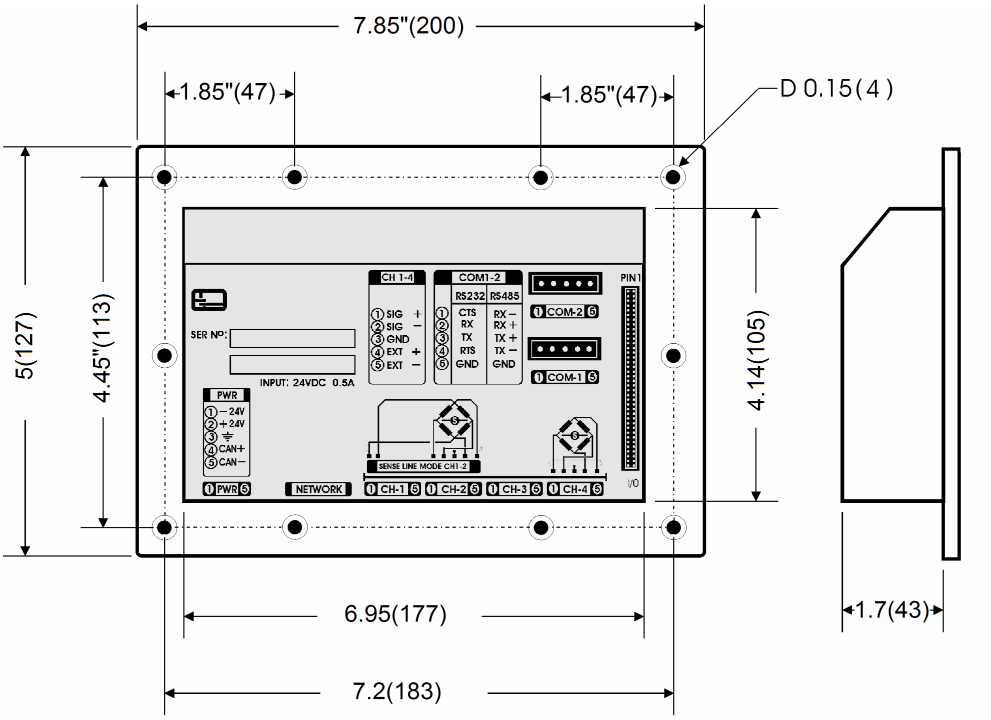
**Observe the following precautions when handling the scale. Once the scale is installed on the idler frame, the weigh assembly becomes a very sensitive measuring instrument. Hitting the weigh idler assembly with a hammer or pressing against the weigh idler forcefully will damage the load cells.**

- Do not pry on the idler, its mountings, or the load cells directly.
- Do not stand or lean on the scale.
- Lift the scale by the weigh blocks only. Do not lift the scale by the idler or idler mounting brackets.
- Never subject the scale to sudden impacts or shocks.

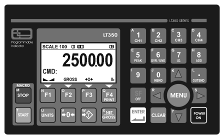
## LT350-BSI BELT SCALE INTEGRATOR - WALL MOUNT NEMA4X

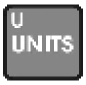


## LT350-BSI BELT SCALE INTEGRATOR - PANEL MOUNT STAINLESS STEEL

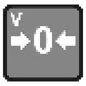




**PANEL CUTOUT (W/H): 6.95 (177) X 4.14(105)**

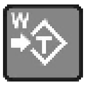





**UNITS** switch between lb and kg. With macros running units cannot be changed. Units can also be locked at startup.

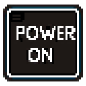



**ZERO** scale can be done with no motion and with the weight within the user programmable zero range. There is also a auto-zero function available.



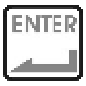

**TARE** scale can be done with a positive weight on the belt scale.


**GROSS or NET** weight display based on a tare entry. The belt scale total can be tared by simply pressing the Tare key and cleared by pressing the CLEAR key




**POWER ON** feature is initially disabled – please see parameter 51 for more detail on how to enable this function.


**MENU** key is the main parameter navigation key. Once selected the user can use the up/down keys to navigate through all the parameters followed by an ENTER key and possibly a password.

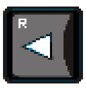



**ENTER** key will execute last command. Use the menu navigation keys to select a parameter followed by the **ENTER** key


**CLEAR** key can be used to clear entry field or exit the menu mode. To clear current tare entry for belt scale.

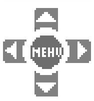
Menu navigation key (single line scroll) and numeric up dial.

Menu navigation key (single line scroll) and numeric down dial.



Menu navigation key or speed scroll – up.

Menu navigation key or speed scroll – down.



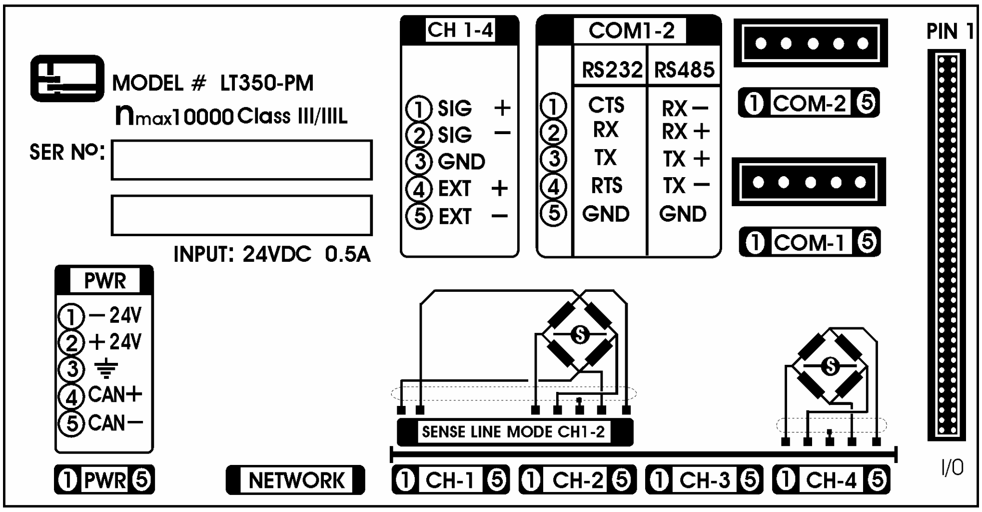
Menu navigation keys - used to select a menu item and numeric dialing.


**SF** special function key used to switch between alphanumeric key entry

## SCALE DISPLAY INDICATORS


| | |
|---|---|
| BELT SCALE 99 | Belt Scale network ID. If  remote control is enabled this will be used to communicate with the unit. |
| | Remote IO run state. If remote IO control is enabled, the START and STOP keys will control the run state of the system. |
| | Indicates remote  IO control is enabled active or idle |
| | Indicates  that the special function  key **SF** was pressed to select either lower or upper case characters for user input where applicable |
| | Indicate Weight stabilized |
| NET/GROSS | Indicate weigh mode  NET or GROSS.  A tare weight must be set. |
| →0← | Indicate scale at zero weight  - center of zero |
| kg/lb | Indicate current units selected  either kilograms or pounds |
| [4] | Indicates currently selected  belt scale (only  used in the LT350-BS4  four channel version) |
| CAL | This icon indicates that the user changed calibration parameters and to save those parameters the user should end calibration by pressing the **F1** key. |

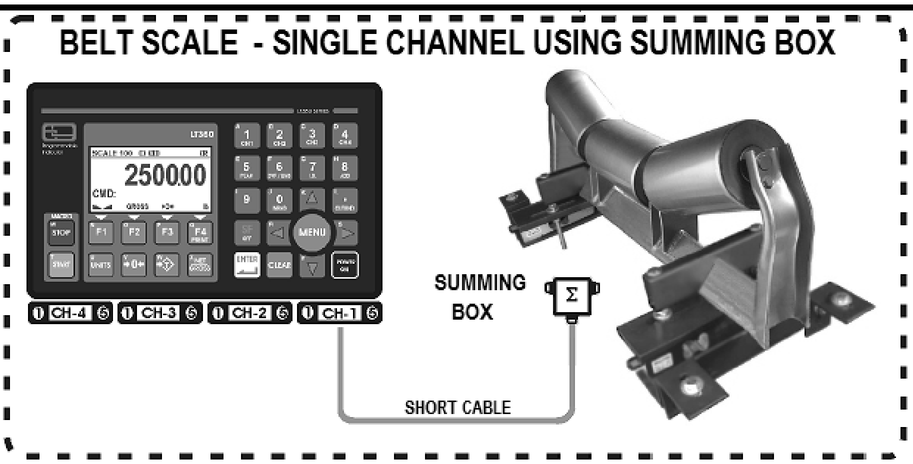
## LT350 HARDWARE INTERFACE DIAGRAM




| | |
|---|---|
| 1 PWR 5 | Power supply input. The indicator accepts a dc voltage between[16-48]Volts. If PoE is used, do not connect power here. Nominal power input should be 24Vdc @ 200mA |
| NETWORK | The controller supports three types of network configurations. Ethernet, Wi-Fi or USB. The Ethernet option also supports PoE (power over Ethernet). |
| SENSE LINE MODE CH1-2 | The sense line mode is a special 6 wire load cell configuration. This configuration uses CH-1 as the sense lines and CH-2 as the weigh idler load cell inputs. This configuration is typically used to improve signal quality over long cable hauls. Command[93] should be used to enable sense line activation. |
| 1 CH-1 5 | The indicator is a single scale multiple weigh idler load cell system that supports between [1-4] weigh idler channels using command[10] |
| 1 COM-1 5 | The primary serial port can be configured as RS232 or RS422 and supports printing or strings using the ASCII.NET protocol |
| 1 COM-2 5 | The primary serial port can be configured as RS232 or RS422 and supports printing or strings using the ASCII.NET protocol |

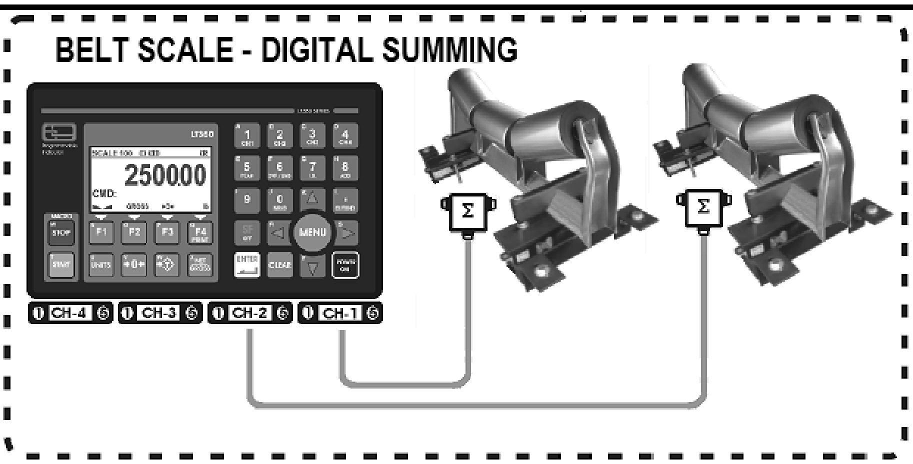
## LT350-BSI-1 SINGLE CHANNEL BELT SCALE CONFIGURATION OPTIONS

The LT350-BSI-1 single belt scale controller supports up to [4] load cells or weigh idlers. Most user simply use one input with a summing box attached. However up to 4 independent weigh idlers can be connected directly to the belt scale controller for higher accuracy. The advantage is that each weigh idler has its own diagnostics and since the system automatically sum the total number of weigh idlers, a much stronger signal is obtained with greater accuracy. The LT350-BSI-1 single belt scale controller supports up to 4 weigh idler channels.



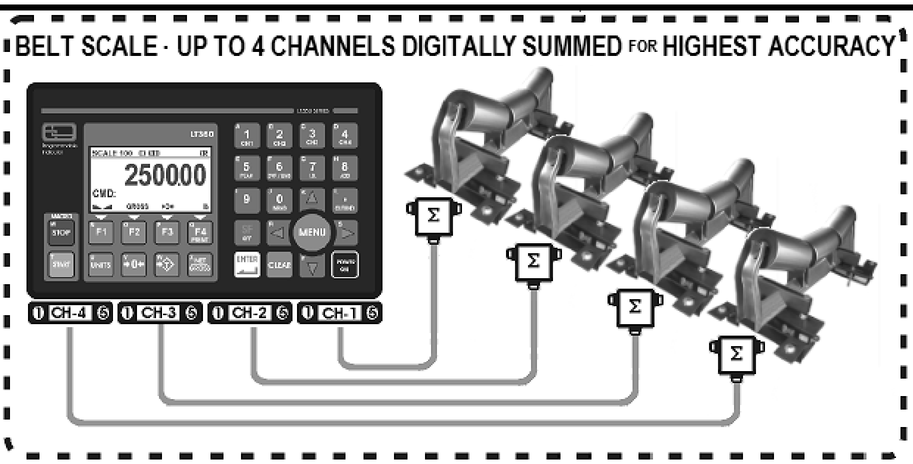

**Advantages**

- Short cable runs
- Easy setup
- Good for summing boxes
- Low cost
- Error < 2%




**Advantages**

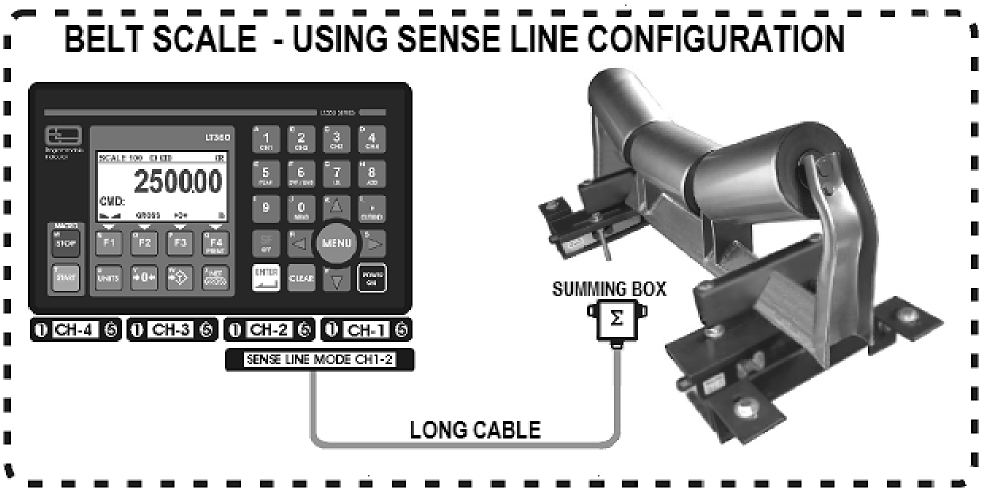
- Digital summing x2
- Digital trimming per idler
- Higher Accuracy
- Strong signal
- Individual diagnostics
- Good signal to noise
- Error < 1.5%

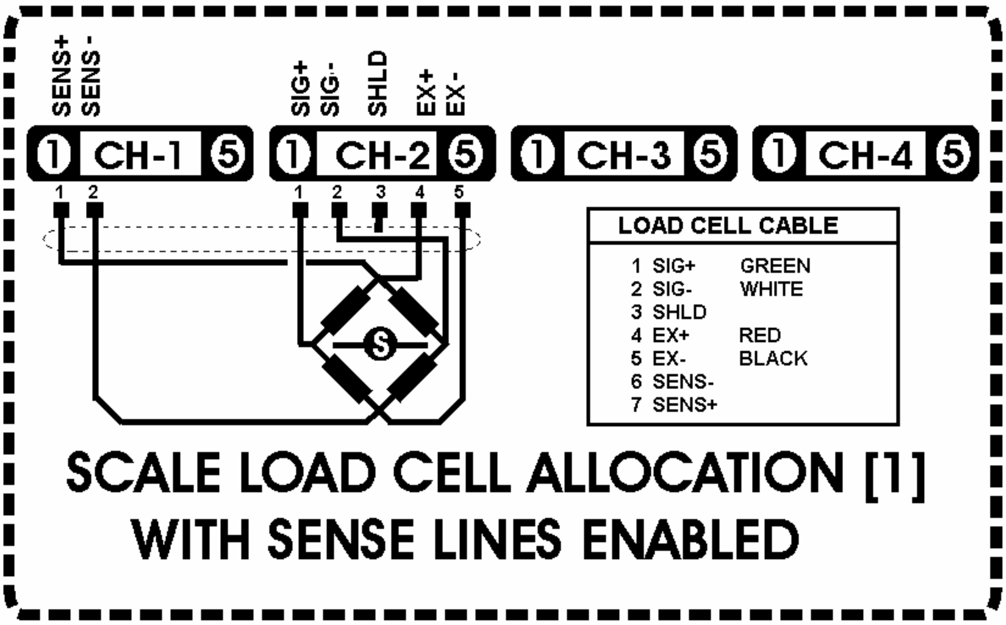



**Advantages**

- Digital summing x4
- Digital trimming per idler
- Very High Accuracy
- Ultra strong signal
- Individual diagnostics
- Excellent signal to noise
- Best accuracy
- Error < 0.5%

If the sense line mode is used **COMMAND[93]**, the CH1 becomes the sense line input while CH2 becomes the weigh idler scale input, while CH3 and CH4 are not used if sense lines are enabled. This configuration is typically used for applications in an electrical noisy environment, or for long cable drops to minimize signal degradation. The user can also place a summing board at the end of the cable to facilitate a multiple weigh idler configuration.

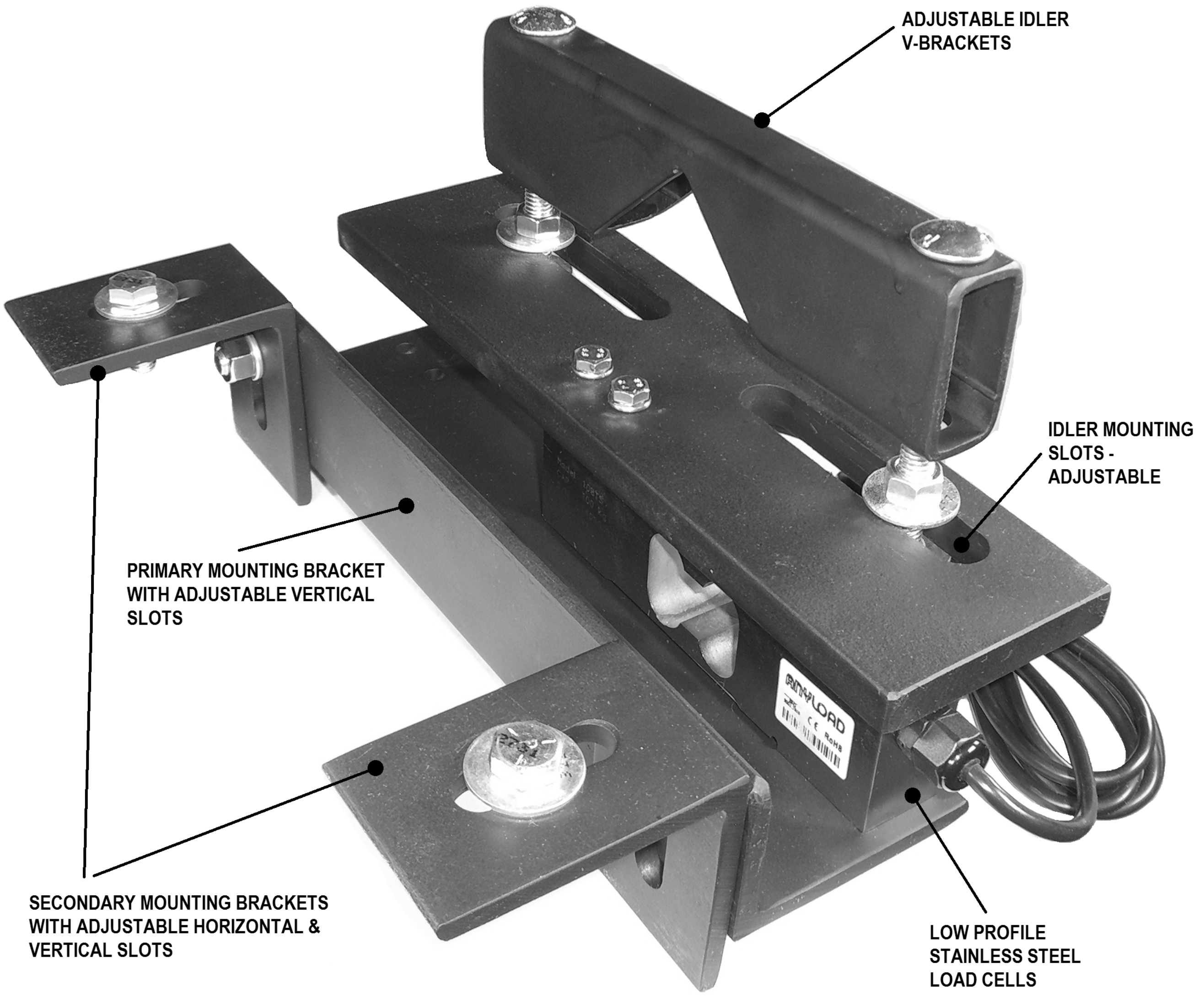


## LT45 – BS SCALE CARRIAGE ASSEMBLY

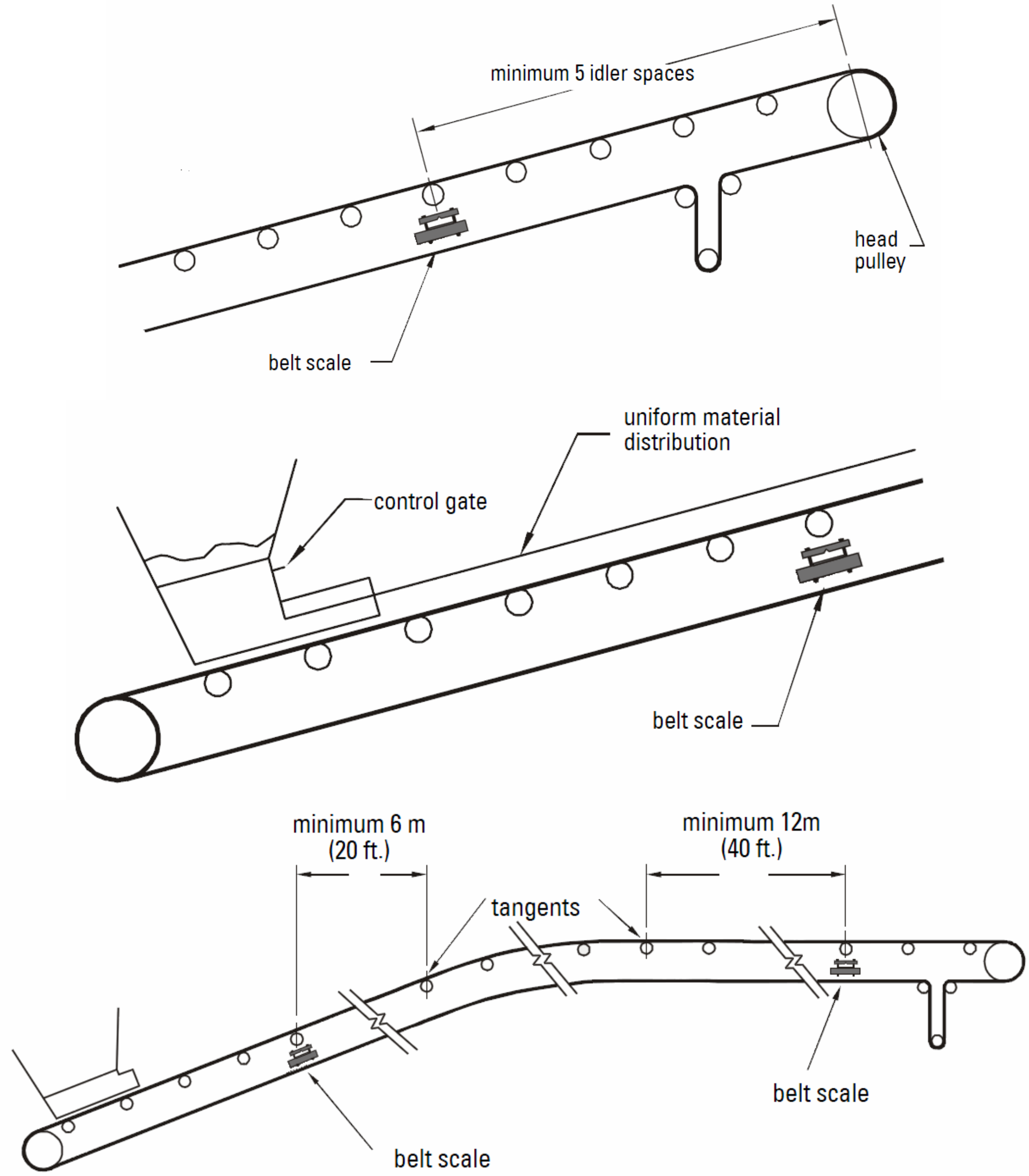
Each weigh idler is supported by two LT45-BS scale modules to make up the scale carriage. The scale bracket assembly is unique in that it does not require any modifications to the idler. The secondary brackets are spaced wide enough to make most types of idlers pass through unhindered. The optional secondary brackets can be removed if not required and the scale module can be mounted using the primary mounting bracket for a extremely low profile solution.

Maximum allowable idler spine for fitting to the LT45-BS  belt scale carriage is:
Angled spine: 75 mm (3.)
Channel spine: 100 mm (4.)



## IDLER SPACING AND LOAD CELL PLACEMENT

The capacity of the belt scale is rated on the maximum continuous load that can be carried across any single weigh idler. The capacity of the conveyor should be known prior to determining the components of the scale system. The load cells should be sized to operate across a loading range with a marginal safety factor. The minimum net loading should be greater than 10% of the rated capacity and the maximum loading should be less than 65%.

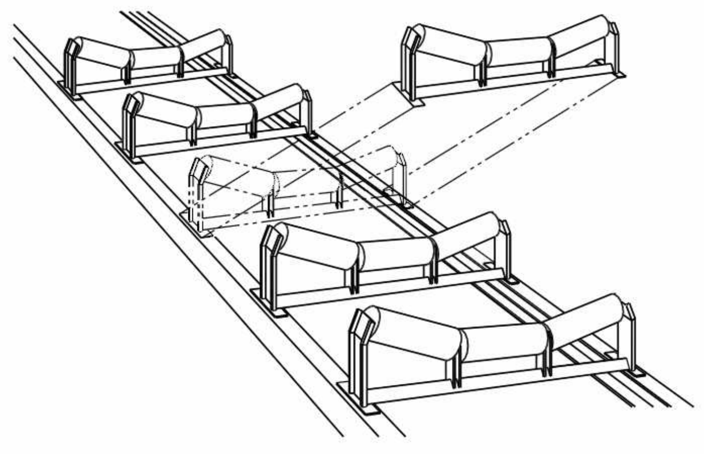


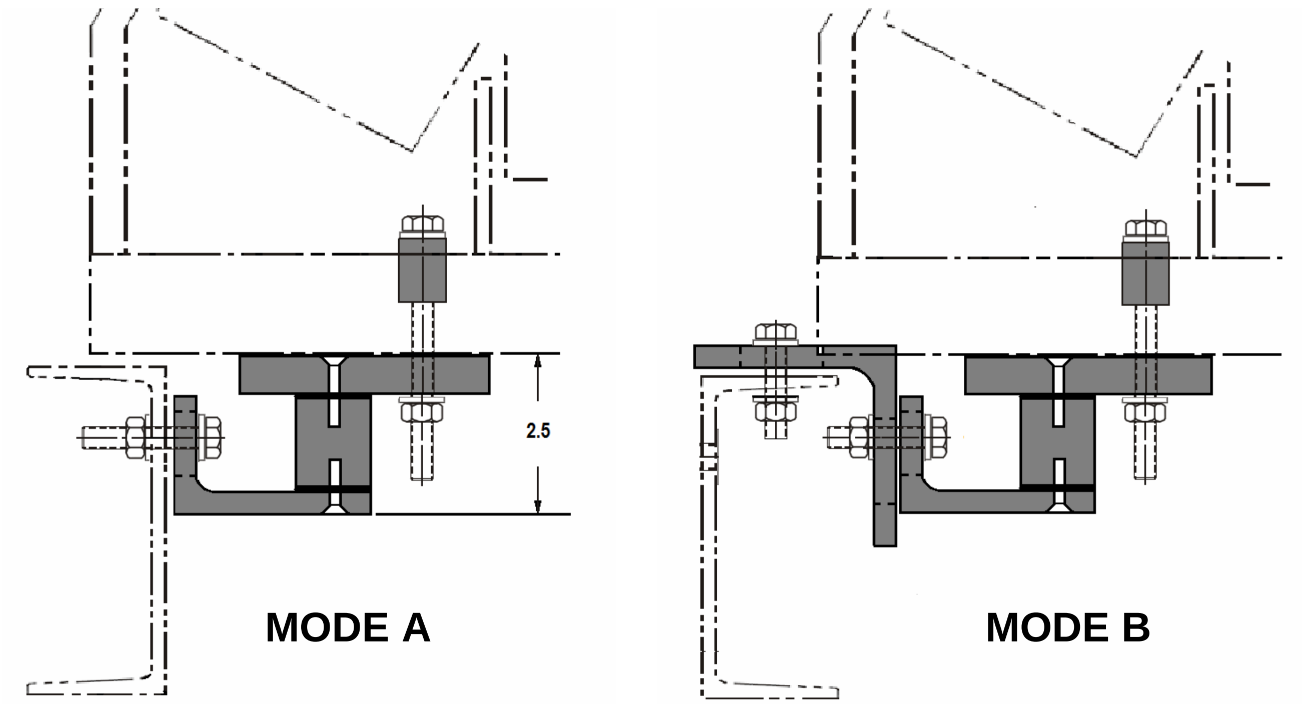
## LT45 – BS SCALE CARRIAGE ASSEMBLY

The LT45-BS belt scale assembly is designed in such a way that requires no modification to the idler frame. The LT45-BS assembly can be mounted in two modes as outlined below – depending on the space requirements and the type of belt scale frame. The LT45-BS belt scale module supports two types of frame mountings as illustrated in diagrams modes A and B.

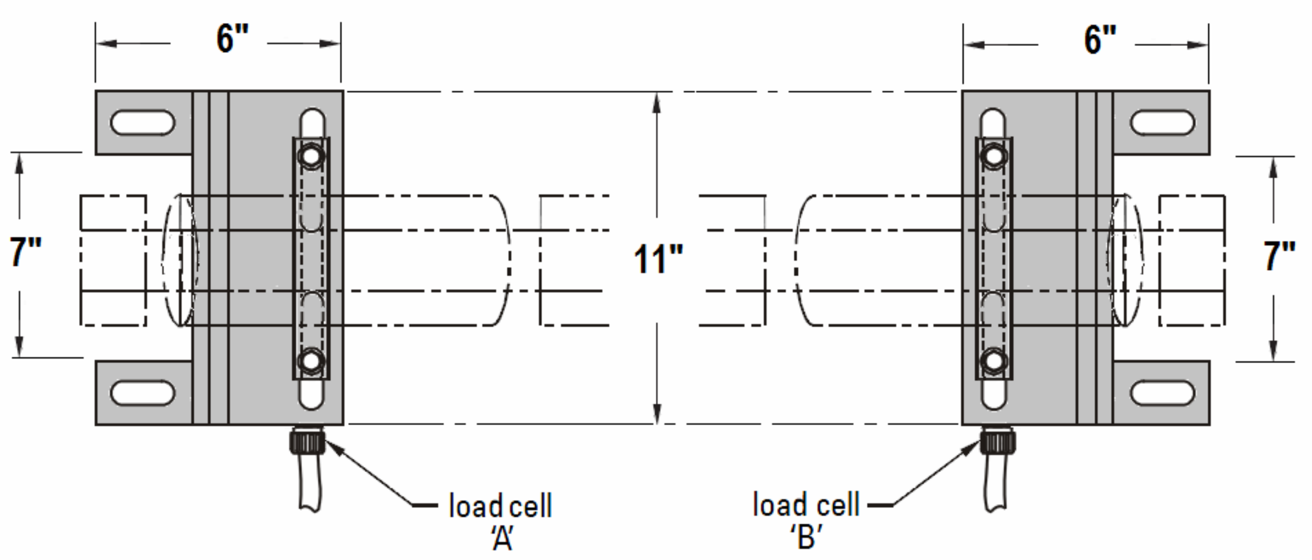
Maximum allowable idler spine for fitting to the LT45-BS belt scale carriage is:

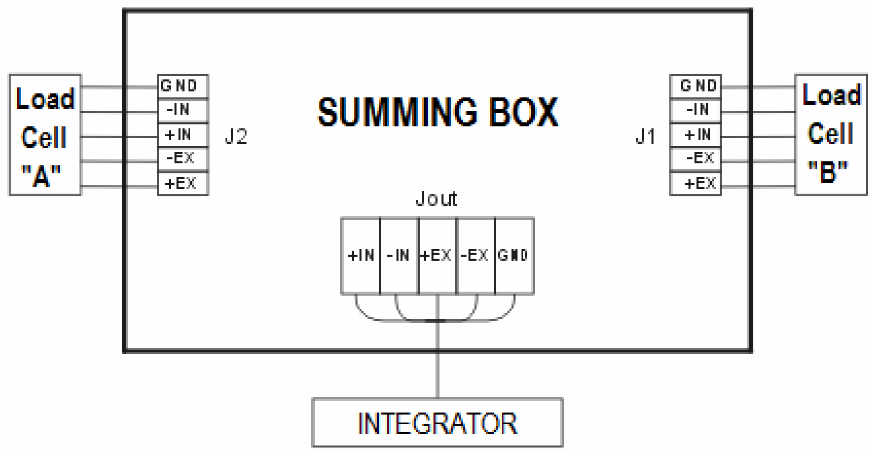
**Angled spine: 75 mm (3.)**
**Channel spine: 100 mm (4.)**





# HOW TO INSTALL AND CONNECT A SUMMING BOX



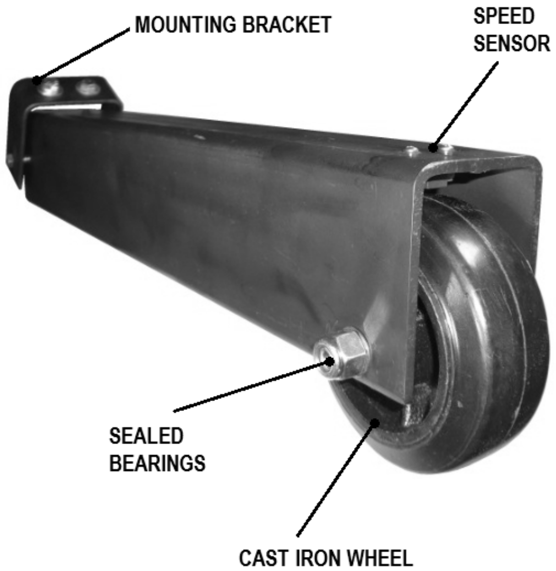



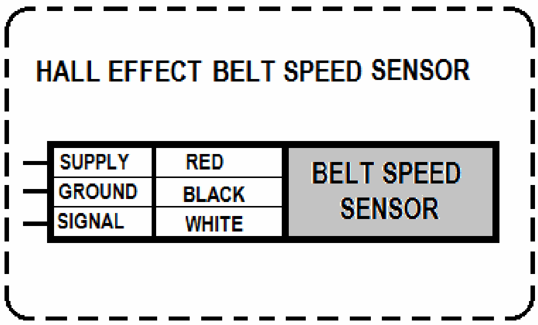
## LT45-SS SCALE SPEED SENSOR ASSEMBLY

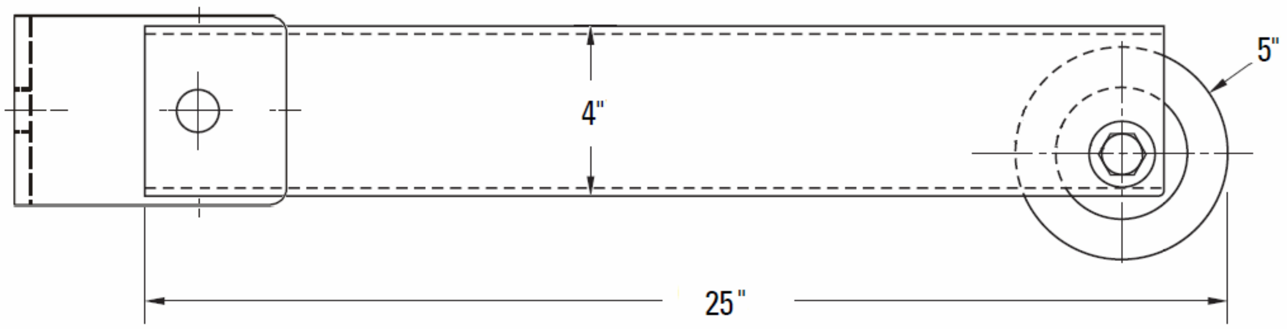
The LT45-BSS belt speed senor assembly has a rugged cast iron rubber wheel with sealed bearings incorporated with a magnetic electronic 3 wire speed sensor that interfaces directly with the LT350-BSI integrator. The mounting bracket can easily be tied to a static idler or cross bar.

The belt speed wheel must maintain constant positive contact between the roll and the belt for proper operation. The speed sensor should never come in contact with material that is being conveyed along the belt nor the belt itself. The signal generated by the speed wheel is converted by the integrator into a value that represents belt travel distance.

The LT350-BSI integrator support both static speed settings or variable belts speeds using the LT45-SS speed sensor assembly. The speed sensor is rated up to 50 feet per second.



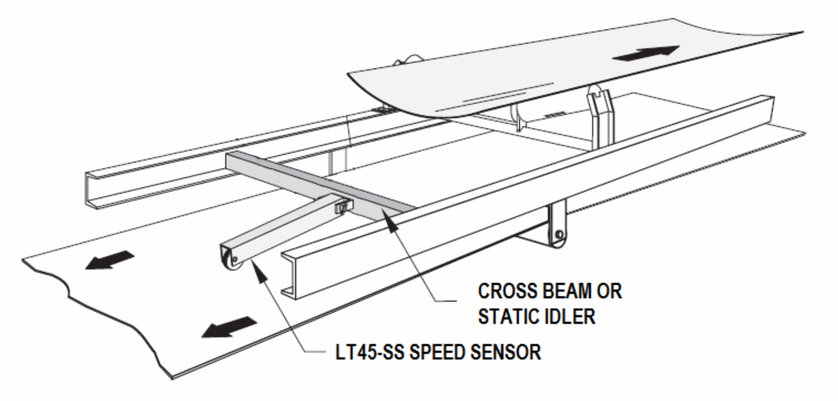



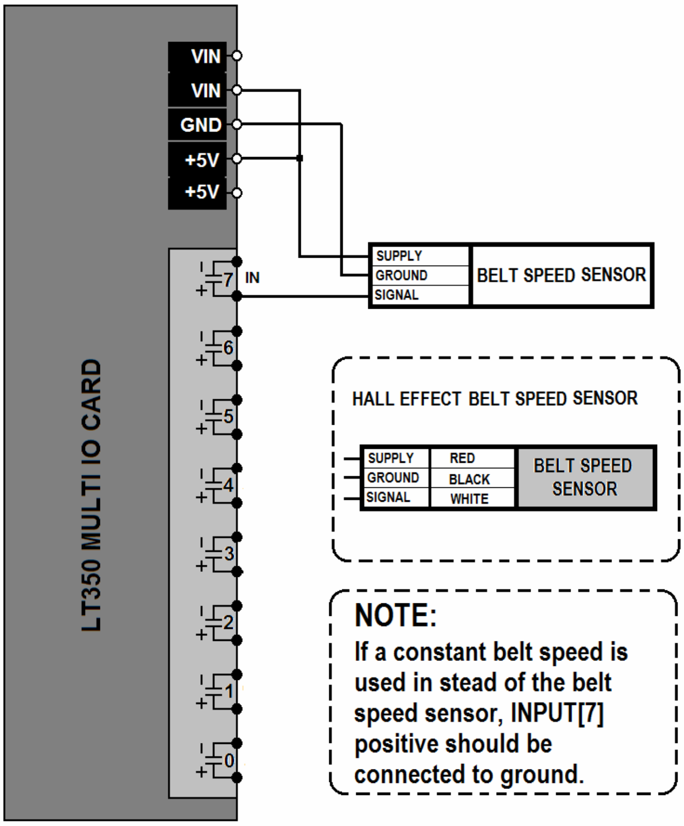



## HOW TO WIRE THE LT45-SS

The LT350-BSI belt scale integrator supports one belt speed pulse sensor input connected to INPUT[7] as outlined in the diagram below. The speed sensor assembly is a 5V, 3 wire device that gets power from the internal 5V supply line on the integrator. Mount the wheel arm to the cross brace closest to the tail of the conveyor. The wheel assembly must be free to move in the vertical direction and must maintain contact with the belt at all times.

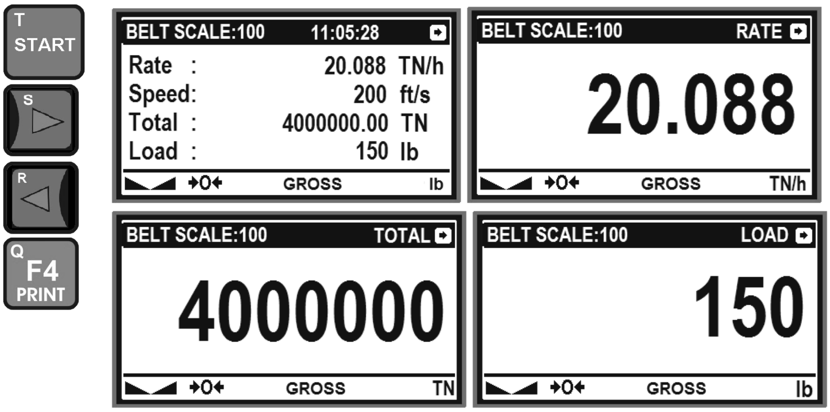
*If a constant speed is selected from the belt scale setup menu items, the user should connect INPUT[7] positive to the ground pin.*

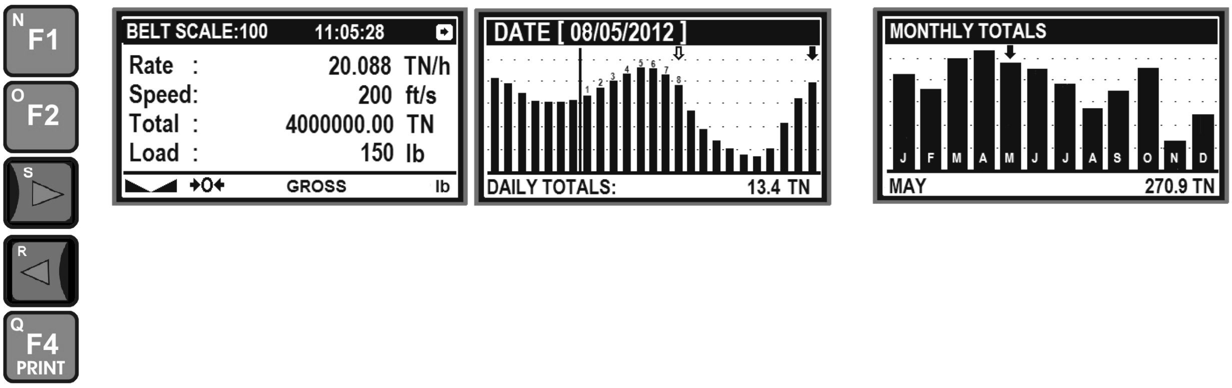


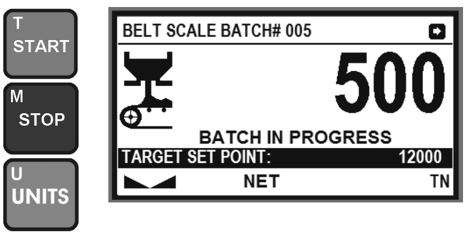



## HOW TO NAVIGATE THE INTEGRATOR MENU FUNCTIONS

The integrator function is activated by pressing **START** and terminated by pressing **STOP** repeatedly.

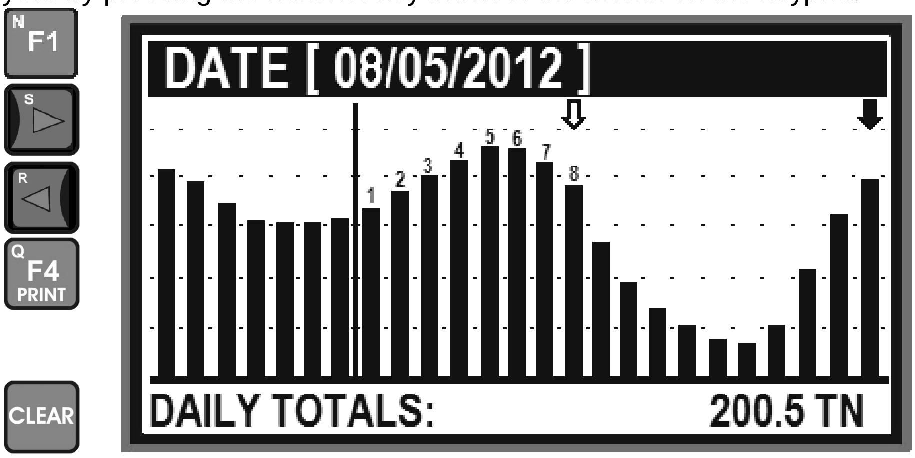


The data log function calculates and stores the daily totals for up to 12 months. The data log function is a very useful function to track productivity and the rate of material moved over up to one year. The data log function must first be enabled by accessing calibration parameter **[89] DATA LOG SETUP**. The data log function only works while the integrator is running **[START].** For best results, the integrator should be permanently powered up, free from power failures and with the integrator running continuously.

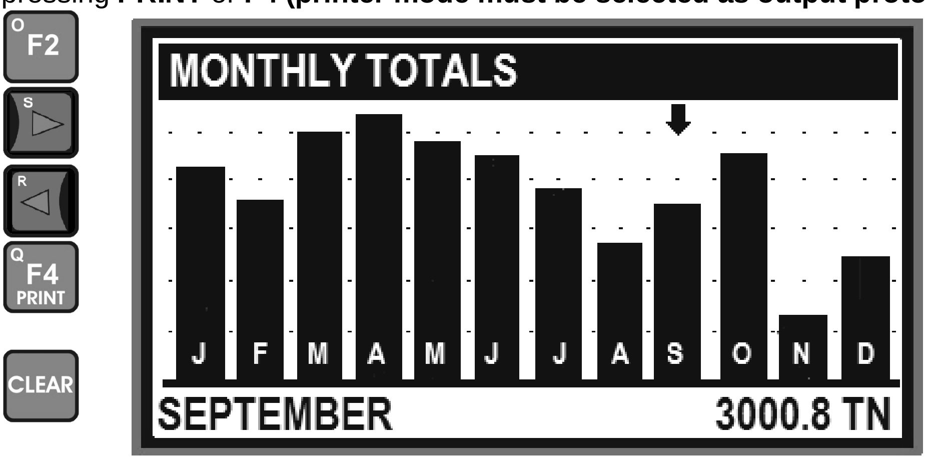
Pressing **F1** from the main display of the integrator will plot in real time the last 30 days of daily totals. The user can print the list of daily totals by pressing **PRINT** or **F4**. The user can view any month of the year by pressing the numeric key index of the month on the keypad.




The user can view the daily totals by moving the arrow to the date of interest. The graph indicates that the daily total for 08/05/2012 was 200.5 TN. New data is always added on the right while older data is shifted to the left. A vertical line indicates the month separator. **Press CLEAR to exit the graphics plot function**

The above function allows the user to scroll and display totals of the last 30 days by using the left and right arrows. While in scroll mode the display will flash the cursor and the date to indicate that the currently displayed daily total is historical and not real time. After a 5 second timeout the display will reset itself to displaying the current daily total at the very right side of the screen. The user can view any month of the year by pressing the numeric key index of the month on the keypad.

Pressing **F2** from the above display will allow the user to view and scroll left or right through each monthly total for the last 12 month as indicated by the cursor. The user can press **ENTER** to view the daily total graph of the currently selected month. The user can print the sum totals for each month by pressing **PRINT** or **F4 (printer mode must be selected as output protocol on the serial port)**

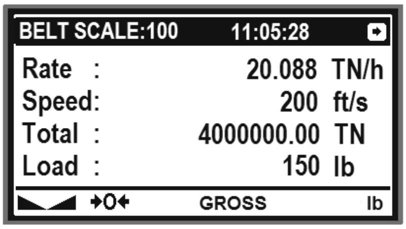


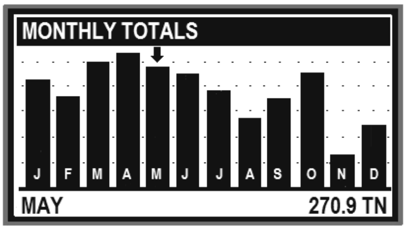

The data view function will display daily totals for all valid entries. Invalid data entries will be displayed as empty fields.

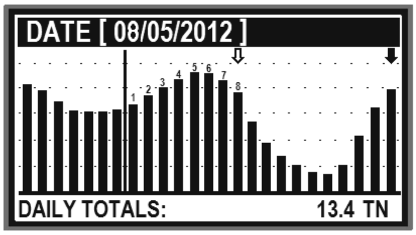
**Press CLEAR to exit the monthly data view function.**

# HOW TO PRINT DATA LOG REPORTS

The data log reports are printed on a serial port that has its protocol set to TICKET PRINTING. The log data can be printed to any RS232 serial printer with a minimum column width of 32 characters or more. The data log function must first be enabled by accessing calibration parameter **[89] DATA LOG SETUP**. The serial port for COM1 or COM2 needs to be configured for TICKET PRINTING and must have the correct baud rate selected. For each display the **PRINT** or **F4** key can be pressed to request a report.




DATE: 21/05/2013
TIME: 11:05:28

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| | |
|---|---|
| Rate: | 20.2 Tn/h |
| Speed: | 200.5 ft/s |
| Total: | 4 000 000 Tn |
| Load: | 156 lb |

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Daily Total: 12 Tn

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MONTHLY TOTALS (05)

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| | |
|---|---|
| JAN : | 268.3 Tn |
| FEB : | 260.1 Tn |
| MAR : | 272.6 Tn |
| APL : | 300.5 Tn |
| MAY : | 270.9 Tn * |
| JUN : | 263.0 Tn |
| JLY : | 255.6 Tn |
| AUG : | 249.2 Tn |
| SEP : | 254.1 Tn |
| OCT : | 260.5 Tn |
| NOV : | 229.4 Tn |
| DEC : | 241.8 Tn |

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TOTAL: 3216.0 Tn

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DAILY TOTALS MONTH: 05

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| | |
|---|---|
| 01/05/2013 : | 7 012 lb |
| 02/05/2013 : | 10 800 lb |
| 03/05/2013 : | 20 001 lb |
| 04/05/2013 : | 30 001 lb |
| 05/05/2013 : | 65 001 lb |
| 06/05/2013 : | 40 001 lb |
| 07/05/2013 : | 16 200 lb |
| 08/05/2013 : | 13 800 lb |
| 09/05/2013 : | 10 200 lb |
| 10/05/2013 : | 6 012 lb |
| 11/05/2013 : | 2000 lb |
| 12/05/2013 : | 300 lb |
| 13/05/2013 : | 300 lb |
| 14/05/2013 : | 900 lb |
| 15/05/2013 : | 2 600 lb |
| 16/05/2013 : | 4 002 lb |
| 17/05/2013 : | 5 000 lb |
| 18/05/2013 : | 5 012 lb |
| 19/05/2013 : | 10 300 lb |
| 20/05/2013 : | 11 200 lb |
| 21/05/2013 : | 12 200 lb |

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TOTAL: 272 842 lb

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## HOW TO DO A STATIC CALIBRATION OF A BELT SCALE

The belt scale requires two calibration sequences. The first is a static calibration where the system is treated as a normal scale. The second calibration is a material calibration where a known amount of test material is fed on the moving belt scale followed by a correction factor to make the displayed total weight match the test material weight.

*The main strategy is to use a typical working weight equivalent to the weight per weigh idler. So if we have 4 weigh idlers, each idler will carry for example ¼ of the combined weight over 4 idlers. Lets assume ¼ of the weight to be 50 lb. We then take a 50 lb test weight and place it on a single weigh idler while leaving all the other idlers empty. We span the scale and then move the test weight from one weigh idler another to see if all the weigh idlers display the same weight. If not, the weigh idler in question needs to be digitally or analogically tweaked.*

First we discuss the static calibration. This command sequence assumes that all the weigh idler load cells are connected either through individual channels or by using the sense line configuration with the possibility of a summing box. You will have to enter the access password for these parameters. Press **MENU** to access command menu.

*To enter calibration mode, the user needs to exit belt scale mode by pressing the STOP key repeatedly until the user is prompted with the option of exiting belt scale mode into calibration mode.*

- Connect indicator to a stable power source. Do not share cable trays with electric motors and switch gear. Have proper cable shielding in place connected to earth ground at the indicator.
- Connect all weigh idler load cells to the indicator. Do not connect anything to load cell channels not used. Always start with load cell channel 1. Do not skip a channel.
- **COMMAND[10]** – allocate the amount of weigh idlers cells used in this application.
- **COMMAND[11]** – select analog input range for type of load cell used. Most analog load cells use the 20mV range standard.
- **COMMAND[12]** – calibration units can be either lb or kg and should be the same as the test weights used. After calibration is finished, the boot up units used can be selected.
- **COMMAND [15]** – scale capacity is the projected maximum working range of the scale weight. This should not exceed the physical limitations of the structure.
- **COMMAND[16]** – this is the weight allowed above scale capacity before a ***scale over range*** will be displayed. If zero, scale over range will be at scale capacity.
- **COMMAND[17]** – dead load belt scale or zero scale. Before a belt scale can be dead loaded, all weights must be removed from the scale idlers.
- **COMMAND[18]** – to span the scale to the calibrated weights currently on the scale. In order to do this, a single weight of the typical working weight per single weigh idler needs to be placed across the first weigh idler of at least 25% of the scale capacity. After spanning is completed, move the weight (in case of multiple weigh idlers) from weigh idler to weigh idler to see if the weight on the display is still the same. If not, the user need to use COMMAND[19] to digitally tweak the weigh idler channel in question or use potentiometers inside the summing boxes to equalize the weigh readings.
- **COMMAND[19]** – Optional. If this is a multi load cell scale, this interactive command allows the user to tweak each weigh idler or load cell to within a tolerable range. A correction factor is then applied to each load cell or summed weigh idler.
- **COMMAND[20]** – Optional. Zero range. This allows the maximum weight that can be zeroed off when there is no weight on the scale. Default is 2% of scale capacity.
- To save calibration settings, press the **F10** key.

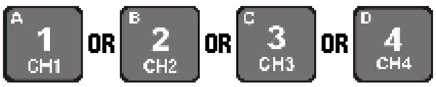
Additional parameters related to **ZERO** and **TARE** settings are optional and should be examined if required.

## LT350-BSI-1 HOW TO PERFORM LOAD CELL ADJUSTMENTS

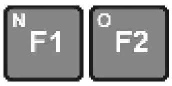
This command makes small adjustments to the span factors of each weigh idler channel traditionally done with analog potentiometers. For example, we have a 2 belt scale idlers that was calibrated to 200lb by placing a weight in the center of each weigh idler. The user then observes that by moving the same weight  from one weigh idler to another, that the weight reading is off by a small margin. To correct for this small margins we use **COMMAND[19].**

- The user executes **COMMAND[19]**.
- The user then selects the weigh idler channel using keys[1,2,3 or 4] to be adjusted.
- The user then places the test weight on the above selected weigh idler.
- The user adjusts the weight using the F3 & F4 function keys until it reads the same as the test weight on the scale.
- The user then moves to the next weigh idler channel using keys **[1,2,3 or 4]**
- If this is not the last weigh idler channel repeat by going to step 3.
- The final step is to press [MENU] to save or abort the changes made.

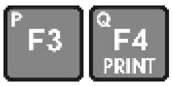
This command is only of use in applications with more than weigh idler channels allocated. Place the test weight as close as possible to the center of the weigh idler that is currently selected on the display.


To select the weigh idler channel to be adjusted

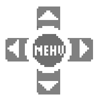



These keys select the coarseness of the adjustments




These keys are used to adjust the weight value for the weigh idler channel in question


Press the this key to exit  without changes



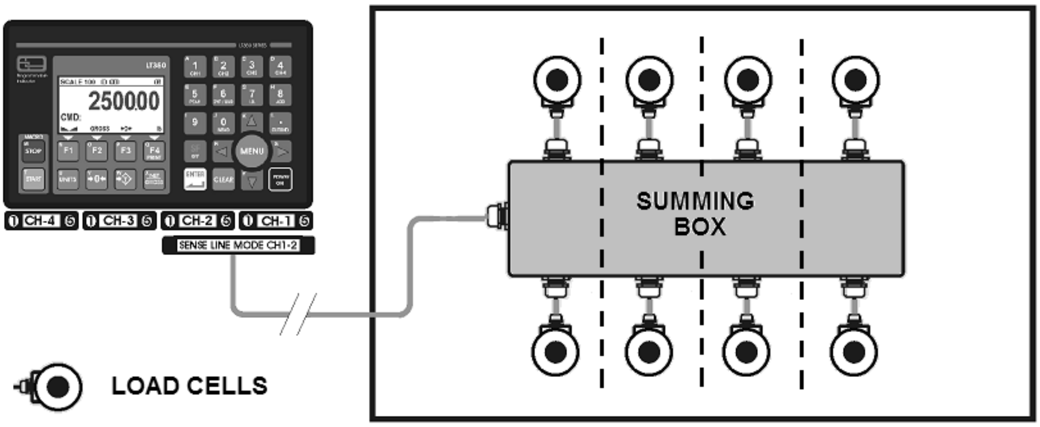

The **[MENU]** key provides the following options:
- Save and exit
- Abort adjustments
- Reset current load cell channel
- Reset all load cell channels
- Return to adjustment menu

## HOW TO TRIM A SUMMING BOX

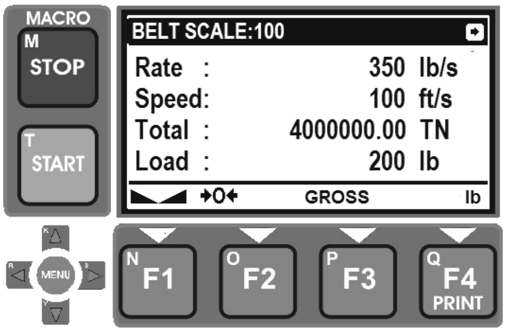
This is a general guideline of connecting multiple weigh idler load cells using a summing box.

Trimming is a process of equalizing the output from multiple individual load cells, or from pairs of cells if using a summing box per weigh idler. When all errors except cell mismatch and cable extensions or reductions have been corrected, continue with the trimming procedures as follows :

1. Set all potentiometers fully counter-clockwise to give maximum signal output from all cells for the weigh idler to be adjusted.
2. Zero the indicator and place calibration test weights over each scale idler in turn. The amount of test weights used depends on the scale configurations; for specific recommendations, refer to Handbook 44 Field Manual. It is recommended to use a weight of at least 25% of the scale capacity not exceeding the maximum capacity of any individual load cell rating.
3. Record the value each time weight is placed on the scale and allow the scale to return to zero to check the friction on other mechanical problems. Select the weigh idler which has the lowest value as your reference point – this weigh idler will not be trimmed.
4. Now repeat the same test weight over each weigh idler and trim each with its potentiometer down to the correct weight. Above sequence might have to be repeated.
5. Tighten all wiring connections and plugs. Pull excess cable out of the enclosure and tighten cord grip assemblies to make it watertight.
6. Connect thick copper rod on outside of metal enclosure to earth ground of the scale chassis for a lightning bypass.



# HOW TO SET INTEGRATOR PARAMETERS




To set belt scale parameters, press STOP to exit to setup mode. Press the MENU key and use the UP/DOWN arrows to select **BELT SCALE SETUP**. These parameters need to be set before attempting a scale or material test calibration.

*Remember to save changes to parameters once finished using F1 once you are back in the main screen.*

## Rate Time Units

This parameter defines what unit of time the rate will be displayed in on the LT350.

- **Hr. - hour (default)**
- Min - minute
- Sec – seconds

## Speed Time Units

This parameter defines how the belt speed is displayed on the LT350

- Hr. - hour
- **Min - minute (default)**
- Sec – seconds

## Measurement Units

The unit of measure parameter defines how the belt is measured. The default is Ft/Min.

- **Ft - feet (default)**
- M – meters

## Rate Units

The unit of rate parameter defines how the rate is displayed.

- **Tn(tonnes) - (default)**
- kg/lb

## Total Units

The integrated weigh total units as displayed.

- **Tn (tonnes) - (default)**
- kg/lb

## Tonnes Type Units

The type of tonnes derived from the base weight unit.

- **TN (short ton) - (default)**
- T (metric ton)
- LT (long ton)

## HOW TO CALIBRATE THE SPEED SENSOR

To set belt speed parameters, press STOP to exit to setup mode. Press the MENU key and use the UP/DOWN arrows to select **BELT SCALE SETUP**. These parameters need to be set before attempting a scale or material test calibration.

*Remember to save changes to parameters once finished using F1 once you are back in the main screen.*

**STEP 1**

Select the belt speed Mode parameter to static(constant speed) or dynamic speed using the LT45-SS belt speed wheel assembly.

**STEP 2**

**Dynamic Speed Calibration:**

Select the belt speed Calibration parameter and while running the belt at a known speed observe the current speed and adjust up or down.

**Static Speed Calibration:**

Select the belt speed Calibration parameter and while running the belt at a know speed observe the current speed and adjust up or down.

| | |
|---|---|
| N F1 | Pressing F1 to decrease incremental factor - fine |
| O F2 | Pressing F2 to increase incremental factor - coarse |
| P F3 | Pressing F3 to increase speed |
| Q F4 PRINT | Pressing F4 to decrease speed |
| MENU | Pressing MENU to save, abort or reset the current adjustments |

*If a constant speed is selected from the belt scale setup menu items, the user should connect INPUT[7] positive to the ground pin. This can also be used for a centrifugal contact sensor that simply indicates belt state. If INPUT[7] is left open the belt speed will be zero in the case of static or constant belt speed mode.*

## HOW TO PERFORM A MATERIAL TEST CALIBRATION

This step is the final weigh calibration following the static weigh calibration procedure as outlined in

HOW TO DO A STATIC CALIBRATION OF A BELT SCALE.

To perform a material test calibration on the totalizer, press STOP repeatedly from the main menu to exit belt mode the setup mode. Press the MENU key and use the UP/DOWN arrows to select **BELT SCALE SETUP** . These parameters need to be set before attempting a scale or material test calibration.

**STEP 1**

Select NUMBER OF IDLERS to select the number of weigh idlers on the conveyor belt.

**STEP 2**

Select the MATERIAL FACTOR menu option using the up/down arrows and press ENTER

**STEP 3**

The MATERIAL FACTOR prompts the user to start the material test once the belt is clear from any material. Use right arrow to select to proceed and press ENTER

**STEP4**

Material of known weight must now be fed on the belt scale once at running speed. Wait for all material to be weighed by the belt scale. Once finished proceed to next step by pressing ENTER

**STEP 5**

The next prompt will ask the user to enter the actual known weight of the test material ( in the base unit of either kg or lb ) in order to calculate the material correction factor.

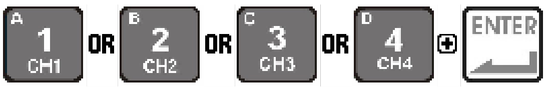
**STEP 6**

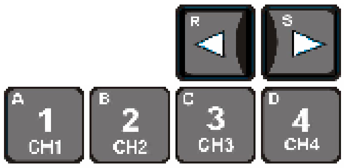
Once the new material factor is calculated, the user will be prompted with the new value for record keeping or for later re-entry. Proceed to finish and save the new factor.

Material test calibration completed.

*Remember to save changes to parameters once finished using F1 once you are back in the main screen.*

Use the [**MENU**] key and up/down keys to navigate to the command or simply type it in as follows:

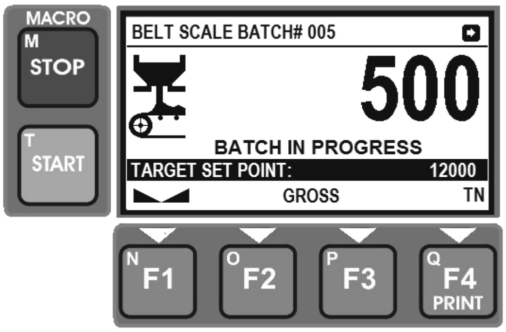



Press these keys to navigate through the weigh idler channel status screens

An alternative method to select the load cell or weigh idler channel

<table>
<tr><th colspan="2">LOAD CELL STATUS INFORMATION</th></tr>
<tr><td>Cell State :</td><td>Runtime diagnostic state of weigh idler or load cell</td></tr>
<tr><td></td><td><table>
<tr><td>PASS</td><td>The weigh idler or load cell , if allocated should be in the [PASS] state</td></tr>
<tr><td>UNUSED</td><td>If a load cell is not allocated using parameter [10] it should show [UNUSED]</td></tr>
<tr><td>FAIL</td><td>If a fault is detected with the load cell it will indicate [FAIL]</td></tr>
</table></td></tr>
<tr><td>Raw Counts:</td><td>Raw counts pertains to the currently selected load cell or summed weigh idler and is the value before converted to a weight quantity. Raw analog to digital counts can be between [0 – 1000 000].</td></tr>
<tr><td>Correction:</td><td>Correction is applied in a belt scale that has more than one weigh idler channels. Due to conveyor scale structural factors it is sometimes required to apply corrections to certain weigh idlers channels by using the correction command [19]. The factory default is [1.000]</td></tr>
<tr><td>Weight:</td><td>Display the current weight across all weigh idlers of the belt scale</td></tr>
<tr><td>Cell/Scale:</td><td>The LT350 is a single scale indicator that can support a belt scale with up to 4 weigh idler channels. The scale allocation command [10] is used to allocate the number of weigh idler channels [1-4]</td></tr>
</table>




To set belt scale batch parameters, press STOP to exit to setup mode. Press the MENU key and use the UP/DOWN arrows to select **BELT BATCH SETUP**. These parameters set warning delay and belt startup timers. Once enabled, press **START** to begin batching using **F3** for the set point.

*Remember to save changes to parameters once finished using **F1** once you are back in the main screen.*

## Enable Batch Mode

This parameter enables batch mode. Once the batch mode is enabled, the user needs to press **F3** to set the weigh set point. The user must then press **START** to activate a batch. If the printer is enabled on **COM1**, batch related data will be printed at the start and end of each batch draft. The batch process can be paused by pressing **STOP**. IO Point [0] controls the feeder bin gate.

## Belt Start-up Timer

This parameter specifies the number of seconds to wait for the belt to come up to speed before opening the material supply gate. The same time period is applied after the set point is reached to give the belt enough time to clear material from the belt. IO Point [1] controls the belt motor.

## Warning Start Timer

This parameter can be used for an alarm to warn that the belt is about to start after the following number of seconds**.** IO Point [2] controls start alarm.

## IO Point[3] External Start Input

This IO point behaves exactly the same as the **START** key on the front panel. Can be wired for remote control.

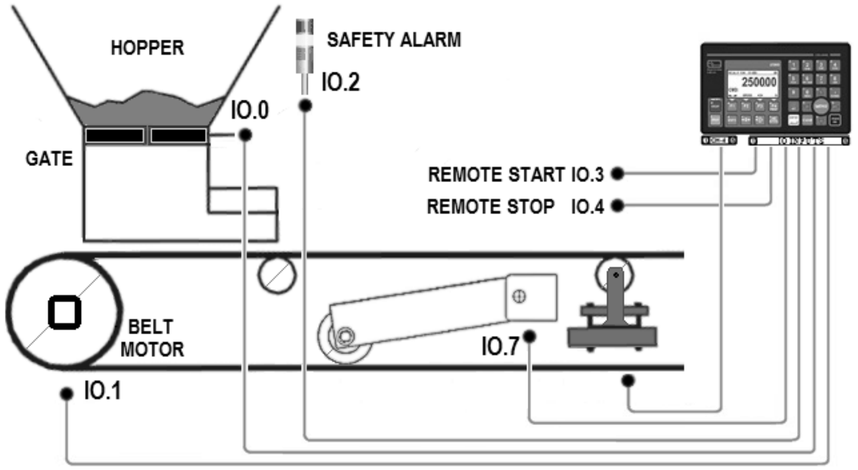
## IO Point[4] External Stop Input

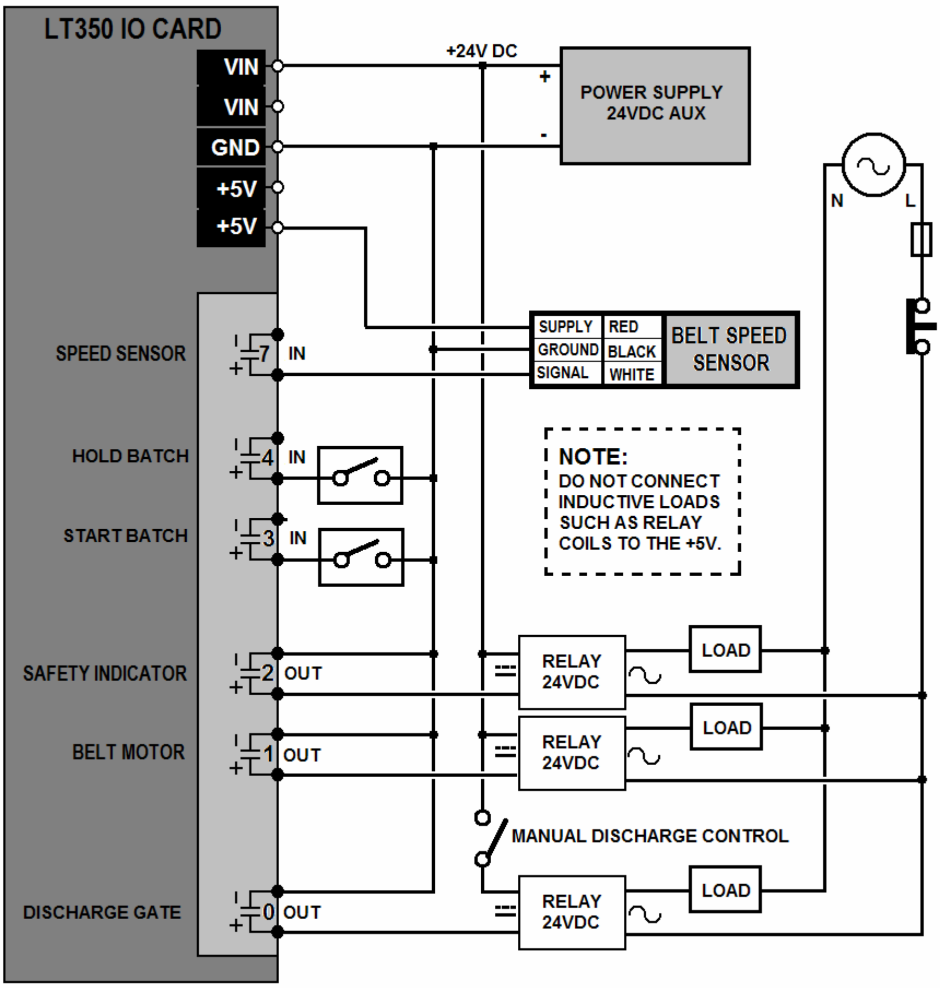
This IO point behaves exactly the same as the **STOP** key on the front panel. Can be wired for remote control.

The batch counter is battery backed up as well as the belt set point which allows the system to recover from power interruptions. In order to reset the batch counter, the user need to enter calibration mode as described earlier on and disable and re-enable the batch function – this will also reset the batch counter. The batch counter is used with the printer reports when the COM port is configured as a printer port.

**F3** is used to enter the set point for the belt batch function. The user can switch between base units lb or kg to Tonnes by pressing the **UNITS** key.

## HOW TO WIRE THE SET POINT INTERFACE



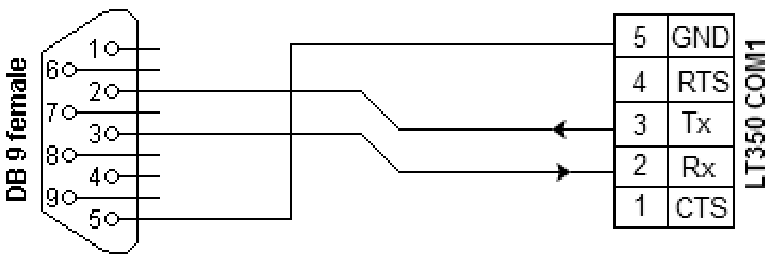
## HOW TO SETUP SERIAL RS232 TO TRANSMIT WEIGHT DATA

To transmit a RS232 ASCII string containing weight information on a serial port do following:

- **COMMAND[34]** – check that the serial port is setup for RS232 mode.
- **COMMAND[35]** – serial baud rate 600,1200,4800,9600,14400,38400,57600 or 230400
- **COMMAND[36]** – parity None, Even or Odd
- **COMMAND[37]** – protocol string format. The LT350 Belt scale data include speed, rate and total related data.
- **COMMAND[39]** – This command dictates whether the string is transmitted automatically or polled via an ASCII command set.
- **COMMAND[52]** – string update rate dictates the pause periods between successive string transmissions where continuous transmission was selected. This is helpful to prevent buffer overruns on the host system.

Windows HyperTerminal can be used to capture serial data strings at the same baud rate

| Computer | LT350 COM1 | Function |
|---|---|---|
| 2 | 2 | Rx ← Tx |
| 3 | 3 | Tx → Rx |
| 5 | 7 | Signal ground |




**NOTE**: *Many PCs today only support RS232 via USB which requires a USB to SERIAL cable*

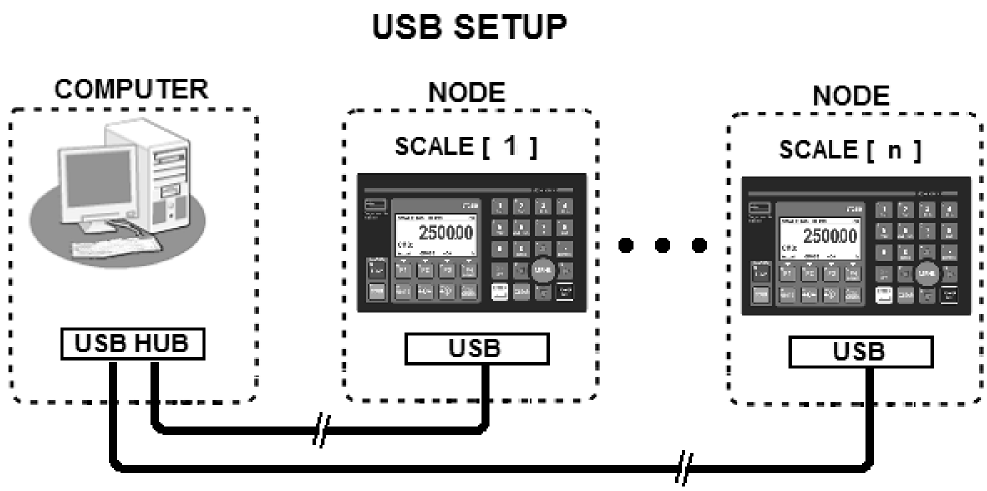
## HOW TO SETUP USB TO TRANSMIT WEIGHT DATA

This setup assumes that a USB port is installed on the LT350. The USB cable can also be the power source to power the LT350. The USB drivers are available on the accompanied CD to support Virtual COM ports on MSWindows platforms. To transmit ASCII strings containing weight information on the USB port of the LT350 do following:

**COMMAND[46]** – Submenu for USB speed and protocol setup

| USB PARAMETERS | |
|---|---|
| | |
| **Data Rate:** | Supported baud rates: 4800, 9600, 14400, 28800, 38400, 57600 or 230400 |
| **Protocol Mode:** | protocol string format. The LT350 Belt scale data include speed, rate and total related data. |
| **String Mode:** | This command dictates whether the string is transmitted automatically or polled via an ASCII command. |

Windows HyperTerminal can be used to capture serial data strings at the same baud rate

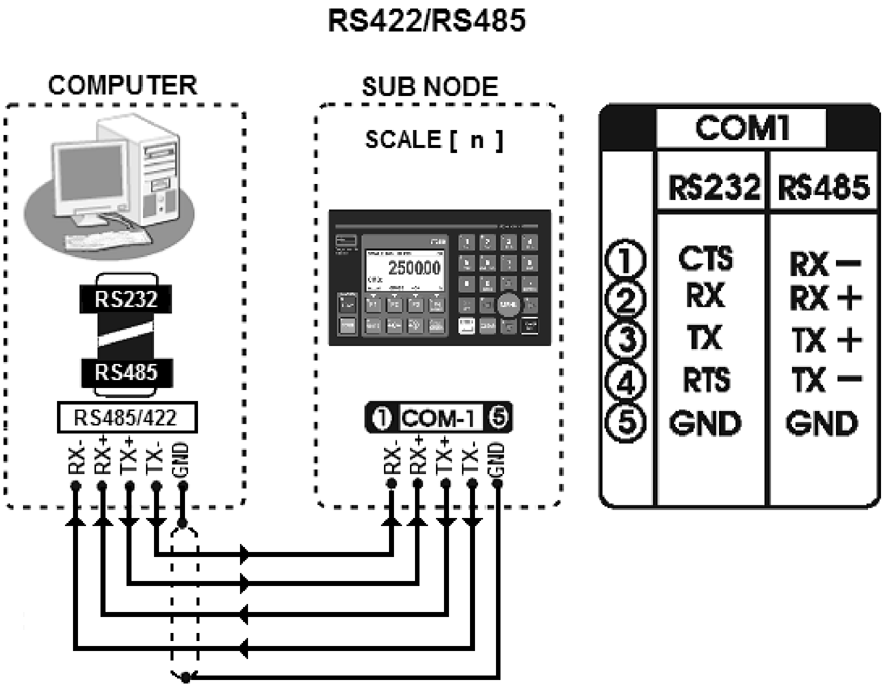


## HOW TO SETUP RS22/485 TO TRANSMIT WEIGHT DATA

To change between RS232 and RS485 mode, the user must first unplug the COM1 cable. This configuration allows for long distance transmission over 100 meters. Unplug serial port, change to RS485 and then do the correct wire setup as outlined below before re-connecting:

- **COMMAND[34]** – Disconnect serial port connector and then set COM1 for RS485 mode. Port may get damaged if a cable is left plugged in that is not an RS485 type.
- **COMMAND[35]** – serial baud rate 600,1200,4800,9600,14400,38400,57600 or 230400
- **COMMAND[36]** – parity None, Even or Odd
- **COMMAND[37]** – protocol string format. The LT350 Belt scale data include speed, rate and total related data.
- **COMMAND[39]** – This command dictates whether the string is transmitted automatically or polled via the ASCII command
- **COMMAND[52]** – string update rate dictates the pause periods between successive string transmissions where continuous transmission was selected. This is helpful to prevent buffer overruns on the host system.

Windows HyperTerminal can be used to capture serial data strings at the same baud rate




**NOTE:** *A RS232 to RS422 converter is required for the computer*

## HOW TO SETUP TCP/IP TO TRANSMIT WEIGHT DATA - I

This setup assumes that a TCP/IP network port is installed on the LT350. The CAT5 network cable can also be the power source to power the LT350 – see PoE:

**COMMAND[47]** – Submenu for internet setup

| INTERNET NETWORK PARAMETERS | |
|---|---|
| **IP address:** | Sets the device IP address when DHCP is off. Factory default is [192.168.1.200]<br>*This address needs to be obtained from the network administrator.* |
| **Subnet Mask:** | Sets the device subnet mask address when DHCP is off. Factory default is [255.255.255.000]<br>*This address needs to be obtained from the network administrator.* |
| **Default Gateway:** | Sets the default gateway address. Factory default is [192.168.1.1]<br>*This address needs to be obtained from the network administrator.* |
| **DHCP Mode:** | If a DHCP server is available on you network you can have an IP address allocated by the server – try not to use DHCP where possible.<br>*You should try to obtain an extended lease on the IP address from the network administrator – typically a real life person.* |
| **Protocol Mode:** | protocol string format. The LT350 Belt scale data include speed, rate and total related data. |
| **String Poll Mode:** | This command dictates whether the string is transmitted automatically or polled via the ASCII command set. |
| **Wireless SSID:** | You should set this SSID to the same as your wireless router SSID. The factory SSID is:**[ Connect ]** |
| **Node ID:** | The node ID [1-100] will be set to the value displayed on the display screen e.g. SCALE 100. If you use the discovery software tool to search for network enabled connected nodes on the internet, the node ID will be used to identify your device. This is particularly important if you will be using DHCP. |

Windows HyperTerminal can be used by selecting WinSock to capture the data strings by specifying the IP address at Port 502

*(continued on next page)*

## HOW TO SETUP TCP/IP TO TRANSMIT WEIGHT DATA II

It is highly recommended to use a separate intelligent Ethernet Switching Router that learns and only forwards data between the two parties that communicate, instead of broadcasting it all across the entire network, slowing everything down – do not use a simple repeating hub.

To setup a LT350 indicator using TCP/IP over CAT5 cable and an Ethernet router, using **COMMAND[47]** Submenu for internet setup.

- Get a static IP address for your LT350 and enter it by selecting IP address in the sub menu in the form of [196.168.200].
- Get the Network mask for the network you connecting to – enter it at the sub menu, Network Mask. Most networks use the default of [255.255.255.000]. This information is readily available on any computer connected to the network – simply copy this information. This information is also available in the network router setup pages.
- Get the Default Gateway information and enter it at the sub menu, Default Gateway. This information should be the same as any computer connected to the network or on the network router.
- DHCP for industrial purposes should be avoided. If the network administrator does not allow static IP addresses for some reason. Try to negotiate a "long lease" for the DHCP assignment.

Alternatively, connect to the indicator through an Ethernet Switch using a standard CAT5 cable (never connect the indicator direct to the PC) web interface to setup the network address information.

- The network setup interface requires a login and password. The default login and password is as follows: LOGIN = **[root]** and PASSWORD = **[dbps]**
- You will then be directed to the TCP/IP address settings. Fill in the IP address, Network Mask and Default Gateway.
- Save the settings and request the LT350 to reboot with new settings – may take a long time, up to 20 seconds for reboot.
- The green LED (furthest from the display) must flash during boot up and then remain off during normal operation.
- The yellow LED (closest to the display) should stay on as soon as a CAT5 cable is connected. It should flicker during data transmissions.

## HOW TO SETUP TCP/IP WiFi TO TRANSMIT WEIGHT DATA

To setup a LT350 indicator using wireless WiFi802.11b, using a Ethernet wireless router or access point, using **COMMAND[47]** Submenu for internet setup:

- Get a static IP address for your LT350 and enter it by selecting IP address in the sub menu in the form of [196.168.200]
- Get the Network mask for the network you are connecting to – enter it at the sub menu, Network Mask. Most networks use the default of [255.255.255.000]. This information is readily available on any computer connected to the network – simply copy this information. This information is also available in the network router setup pages.
- Get the Default Gateway information and enter it at the sub menu, Default Gateway. This information should be the same for any computer connected to the network or on the network router.
- Setup the SSID to be the same name as the access point or wireless router/switch.
- It is best to first get the basics working before enabling advanced WiFi encryption. Once you can communicate with the LT350 across a network. Use the Web interface described below to setup encryption if you need it.

**Alternatively, temporarily change the SSID of the router to the factory default SSID of the LT350, which is [Connect].** You should now be able to find the indicator using the LTDISCOVER.EXE program on the CD. Also temporarily disable encryption on the router until you have the basic communications working. Then use the Web interface to setup the network address information.

- The network setup interface requires a login and password. The default login and password is as follows: LOGIN = **[root]** and PASSWORD = **[dbps]**
- You will then be directed to the TCP/IP address settings. Fill in the IP address, Network Mask, Default Gateway, SSID and enable encryption if needed. The SSID should be the same as the wireless router/switch. Connect the antenna, with a clear line of sight and positioned away from metal structures wherever possible.
- Save the settings and request the LT350 to reboot with new settings – may take up to 10 seconds for reboot.
- The green LED (furthest from the display) must flash during boot up and then remain off during normal operation.
- The yellow LED (closest to the display) should stay on as soon as a CAT5 cable is connected. It should flicker during data transmissions.

**[<][C][T][TTTTTTTTTTTTT][,][U][:][D][>][C][C]**

| | | | | |
|---|---|---|---|---|
| 0<br>**[<]** | 1<br>**[C]** | | | *(belt number)* |
| 2<br>**[:]** | 3<br>**[T]** | 15<br>**[,]** | 16<br>**[U]** | *(belt Total)* |
| 17<br>**[:]** | 18<br>**[D]** | | | *(belt status)* |
| 19<br>**[>]** | 20<br>**[C]** | 21<br>**[C]** | | *(belt checksum)* |

| | | | |
|---|---|---|---|
| **0** | **<** | 1 byte | String start delimiter |
| **1** | **C** | 1 byte | Belt scale channel number [1-4] |
| **2** | **T** | 1 byte | Parameter start delimiter |
| **3** | **T** | 13 bytes | **Belt Total – 13 digits with one decimal point, fixed.** |
| **4** | **,** | 1 byte | Parameter separator |
| **17** | **U** | 1 byte | Total units: L(lb), K(kg) or T(tonnes) |
| **18** | **:** | 1 byte | Parameter start delimiter |
| **19** | **D** | 1 byte | System state Diagnostics |
| **20** | **>** | 1 byte | String end delimiter |
| **21** | **C** | 1 byte | Upper checksum hex byte for inverted checksum of bytes 0-21 |
| **22** | **C** | 1 byte | Lower checksum hex byte for inverted checksum of bytes 0-21 |

System state Diagnostics (row 19):

| | |
|---|---|
| **[SPACE]** | Indicates normal operation |
| **O** | Indicates scale over capacity |
| **N** | Indicates scale negative |
| **M** | Indicates motion on scale |
| **Z** | Scale at zero weight [Z] or not at zero [SPACE] |
| **H** | Integrator halted |

**[<][C][T][TTTTTTTTTTTTT][,][U][S][SSSSSSSS][,][U][:][D][>][C][C]**

| | | | | |
|---|---|---|---|---|
| 0 [<] | 1 [C] | | | *(belt number)* |
| 2 [T] | 3 [T] | 15 [,] | 16 [U] | *(belt total)* |
| 17 [S] | 18 [S] | 26 [,] | 27 [U] | *(belt speed)* |
| 28 [:] | 29 [D] | | | *(belt state )* |
| 30 [>] | 31 [C] | 32 [C] | | *(checksum)* |

| | | | |
|---|---|---|---|
| **0** | **<** | 1 byte | String start delimiter |
| **1** | **C** | 1 byte | Belt scale channel number [1-4] |
| **2** | **T** | 1 byte | Parameter start delimiter |
| **3** | **T** | 13 bytes | **Belt Total – 13 digits with one decimal point, fixed.** |
| **15** | **,** | 1 byte | Parameter separator |
| **16** | **U** | 1 byte | Total units: L(lb), K(kg) or T(tonnes) |
| **17** | **S** | 1 byte | Parameter start delimiter |
| **18** | **S** | 8 bytes | **Belt speed** |
| **26** | **,** | 1 byte | Parameter separator |
| **27** | **U** | 1byte | Belt speed units: f**t/s, ft/m, ft/h, M/s, M/m, M/h** |
| **28** | **:** | 1 byte | Parameter start delimiter |
| **29** | **D** | 1 byte | Belt State<br>**[SPACE]** Indicates normal operation<br>**O** Indicates scale over capacity<br>**N** Indicates scale negative<br>**M** Indicates motion on scale<br>**Z** Scale at zero weight [Z] or not at zero [SPACE]<br>**H** Integrator halted |
| **30** | **>** | 1 byte | String end delimiter |
| **31** | **C** | 1 byte | Upper checksum hex byte for inverted checksum of data |
| **32** | **C** | 1 byte | Lower checksum hex byte for inverted checksum of data |

**[<][C][T][TTTTTTTTTTTTT][,][U][S][SSSSSSSS][,][U][R][RRRRRRRR][,][U][:][D][>][C][C]**

| | | | | |
|---|---|---|---|---|
| 0<br>**[<]** | 1<br>**[C]** | | | *(belt number)* |
| 2<br>**[T]** | 3<br>**[T]** | 15<br>**[,]** | 16<br>**[U]** | *(belt total)* |
| 17<br>**[S]** | 18<br>**[S]** | 26<br>**[,]** | 27<br>**[U]** | *(belt speed)* |
| 28<br>**[R]** | 29<br>**[R]** | 37<br>**[,]** | 38<br>**[U]** | *(belt rate)* |
| 39<br>**[:]** | 40<br>**[D]** | | | *(belt state )* |
| 41<br>**[>]** | 42<br>**[C]** | 43<br>**[C]** | | *(checksum)* |

| | | | |
|---|---|---|---|
| **0** | **<** | 1 byte | String start delimiter |
| **1** | **C** | 1 byte | Belt scale channel number [1-4] |
| **2** | **T** | 1 byte | Parameter start delimiter |
| **3** | **T** | 13 bytes | **Belt Total – 13 digits with one decimal point, fixed.** |
| **15** | **,** | 1 byte | Parameter separator |
| **16** | **U** | 1 byte | Total units: L(lb), K(kg) or T(tonnes) |
| **17** | **S** | 1 byte | Parameter start delimiter |
| **18** | **S** | 8 bytes | **Belt speed** |
| **26** | **,** | 1 byte | Parameter separator |
| **27** | **U** | 1byte | Belt speed units: f**t/s, ft/m, ft/h, M/s, M/m, M/h** |
| **28** | **R** | 1 byte | Parameter start delimiter |
| **29** | **R** | 8 bytes | **Belt rate** |
| **37** | **,** | 1 byte | Parameter separator |
| **38** | **U** | 1byte | Belt speed units: **kg**/s/m/h, **lb**/s/m/h, **T**/s/m/h |
| **39** | **:** | 1 byte | Parameter start delimiter |
| **40** | **D** | 1 byte | Belt State<br>**[SPACE]** Indicates normal operation<br>**O** Indicates scale over capacity<br>**N** Indicates scale negative<br>**M** Indicates motion on scale<br>**Z** Scale at zero weight [Z] or not at zero [SPACE]<br>**H** Integrator halted |
| **41** | **>** | 1 byte | String end delimiter |
| **42** | **C** | 1 byte | Upper checksum hex byte for inverted checksum of data |
| **43** | **C** | 1 byte | Lower checksum hex byte for inverted checksum of data |

**[<][C][T][TTTTTTTTTTTTT][,][U][S][SSSSSSSS][,][U][R][RRRRRRRR][,][U][L][LLLLLLLL][,][U][:][D][>][C][C]**

| | | | | |
|---|---|---|---|---|
| 0<br>**[<]** | 1<br>**[C]** | | | *(belt number)* |
| 2<br>**[T]** | 3<br>**[T]** | 15<br>**[,]** | 16<br>**[U]** | *(belt total)* |
| 17<br>**[S]** | 18<br>**[S]** | 26<br>**[,]** | 27<br>**[U]** | *(belt speed)* |
| 28<br>**[R]** | 29<br>**[R]** | 37<br>**[,]** | 38<br>**[U]** | *(belt rate)* |
| 39<br>**[L]** | 40<br>**[L]** | 48<br>**[,]** | 49<br>**[U]** | *(belt load)* |
| 50<br>**[:]** | 51<br>**[D]** | | | *(belt state )* |
| 52<br>**[>]** | 53<br>**[C]** | 54<br>**[C]** | | *(checksum)* |

| | | | |
|---|---|---|---|
| **0** | **<** | 1byte | String start delimiter |
| **1** | **C** | 1 byte | Belt scale channel number [1-4] |
| **2** | **T** | 1 byte | Parameter start delimiter |
| **3** | **T** | 13 bytes | **Belt Total – 13 digits with one decimal point, fixed.** |
| **15** | **,** | 1 byte | Parameter separator |
| **16** | **U** | 1 byte | Total units: L(lb), K(kg) or T(tonnes) |
| **17** | **S** | 1 byte | Parameter start delimiter |
| **18** | **S** | 8 bytes | **Belt speed** |
| **26** | **,** | 1 byte | Parameter separator |
| **27** | **U** | 1byte | Belt speed units: f**t/s, ft/m, ft/h, M/s, M/m, M/h** |
| **28** | **R** | 1 byte | Parameter start delimiter |
| **29** | **R** | 8 bytes | **Belt rate** |
| **37** | **,** | 1 byte | Parameter separator |
| **38** | **U** | 1byte | Belt speed units: **kg**/s/m/h, **lb**/s/m/h, **T**/s/m/h |
| **39** | **L** | 1 byte | Parameter start delimiter |
| **40** | **L** | 6 bytes | **Belt Load** |
| **48** | **,** | 1 byte | Parameter separator |
| **49** | **U** | 1byte | Belt Load: kg, lb |
| **50** | **:** | 1 byte | Parameter start delimiter |
| **51** | **D** | 1 byte | **Belt State** |
| **52** | **>** | 1 byte | String end delimiter |
| **53** | **C** | 1 byte | Upper checksum hex byte for inverted checksum of data |
| **54** | **C** | 1 byte | Lower checksum hex byte for inverted checksum of data |

# UNIT CONVERSION TABLE

| Primary Units | Multiplication Factor | Secondary Units | |
|---|---|---|---|
| Pounds (lb) | 0.453592 | kilograms | kg |
| | 0.0005 | Short tons | TN |
| | 0.000446 | Long tons | LT |
| | 0.000453 | Metric tons | T |
| | | | |
| **Primary Units** | **Multiplication Factor** | **Secondary Units** | |
| Kilograms (kg) | 2.20462 | pounds | lb |
| | 0.001102 | Short tons | TN |
| | 0.000984 | Long tons | LT |
| | 0.001000 | Metric tons | T |
| | | | |
| **Primary Units** | **Multiplication Factor** | **Secondary Units** | |
| Short Tons (TN) | 2000.00 | pounds | lb |
| | 907.185 | kilograms | kg |
| | 0.892857 | Long tons | TN |
| | 0.907185 | Metric tons | T |
| | | | |
| **Primary Units** | **Multiplication Factor** | **Secondary Units** | |
| Metric Tons (T) | 2204.62 | pounds | lb |
| | 1000.00 | kilograms | kg |
| | 1.10231 | Short tons | TN |
| | 0.984207 | Long tons | LT |
| | | | |
| **Primary Units** | **Multiplication Factor** | **Secondary Units** | |
| Long Tons (LT) | 2240.00 | pounds | lb |
| | 1016.05 | kilograms | kg |
| | 1.1200 | Short tons | TN |
| | 1.01605 | Metric Tons | T |
| | | | |